

EXERCISE #1

```
#include <iostream>
using namespace std;

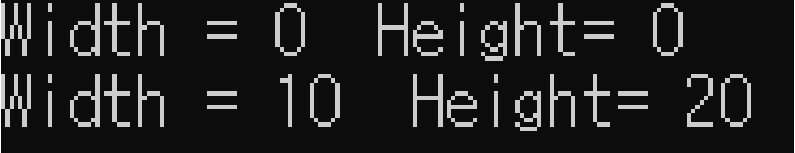
class Rectangle {
    int m_w = 0;
    int m_h = 0;

public:
    Rectangle &setwidth(int a) {
        m_w = a;
        return *this;
    }
    Rectangle &setheight(int b) {
        m_h = b;
        return *this;
    }
    Rectangle &Display() {
        cout << "Width = " << m_w << " Height= " << m_h << endl;
        return *this;
    }
};

int main() {
    Rectangle rect;
    rect.Display();
    rect.setwidth(10).setheight(20).Display();

    return 0;
}
```

❖ This code uses this pointer.



```
Width = 0 Height= 0
Width = 10 Height= 20
```

EXERCISE #2

```
#include <iostream>
using namespace std;

class IntArray {

private:
    int m_len{ 0 };
    int* m_data{ nullptr };

public:
    IntArray(int len): m_len ( len )
    {
        m_data = new int[m_len];
    }
}
```

```

IntArray(IntArray& other) : m_len(other.m_len) {}
~IntArray() {
    if (m_data) delete[] m_data;
}
void set(int index, int value) {
    if (index >= 0 && index < m_len)
        m_data[index] = value;
}
int get(int index, int err) const {
    if (index >= 0 && index < m_len)
        return m_data[index];
    else
        return err;
}
};

int main() {
    int i;

    cout << "=== IntArray a{ 10 } ===" << endl;
    IntArray a{ 10 };
    for (i = 0; i < 10; ++i)
        a.set(i, i * 10 + 5);
    cout << "a: ";
    for (i = 0; i < 10; ++i)
        cout << a.get(i, -1) << ' ';
    cout << endl;

    cout << "=== IntArray b{ a } ===" << endl;
    IntArray b{ a };
    cout << "b: ";
    for (i = 0; i < 10; ++i)
        cout << a.get(i, -1) << ' ';

    return 0;
}

```

- ❖ This code needed copy constructor in order to work so I wrote a copy constructor of IntArray. (IntArray(IntArray& other) : m_len(other.m_len) {} has an copy named other)

```

=== IntArray a{ 10 } ===
a: 5 15 25 35 45 55 65 75 85 95
=== IntArray b{ a } ===
b: 5 15 25 35 45 55 65 75 85 95

```